ANSIBLE_TASK_01

1. Setup one master and two worker nodes in ansible.

Step 1: Create three EC2 instances

Open the AWS Console → EC2 → Instances → Launch instances.

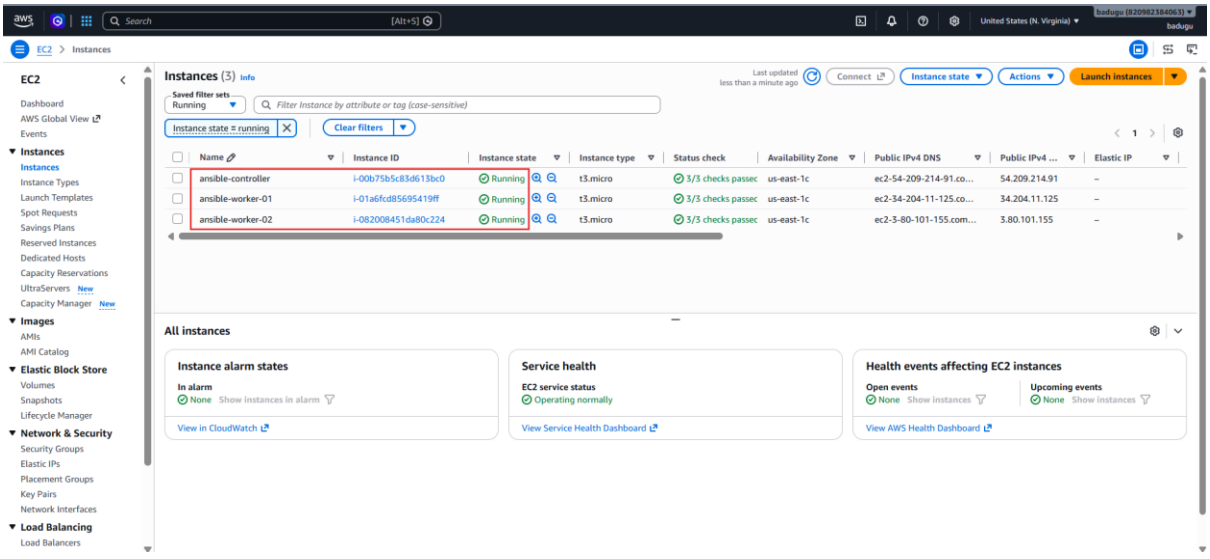
Use these settings for each instance:

Setting	Master	Worker 1	Worker 2
Name	ansible-controller	ansible-worker1	ansible-worker2
OS image	Amazon Linux Server 24.04 LTS	Amazon Linux Server 24.04 LTS	Amazon Linux Server 24.04 LTS
Instance type	t3.micro	t3.micro	t3.micro
Key pair	Create or select one	Same key pair	Same key pair
Storage	8–10 GiB	8–10 GiB	8–10 GiB

Use the same VPC and preferably the same subnet for all three instances.

Security group: Allow SSH (TCP port 22) from your own public IP to the master. For worker access, allow SSH from the master's private IP or its security group. Avoid opening SSH to everyone (0.0.0.0/0) unless temporarily necessary.

Wait until all three instances show Running and their status checks pass.

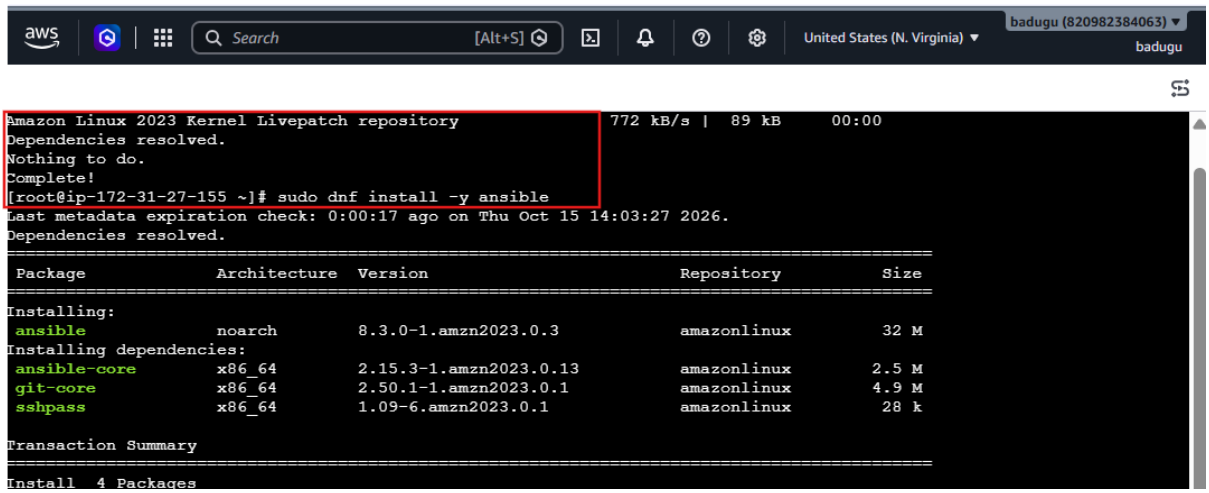


Step 2: Connect to the Ansible Master and Install Ansible on the Master

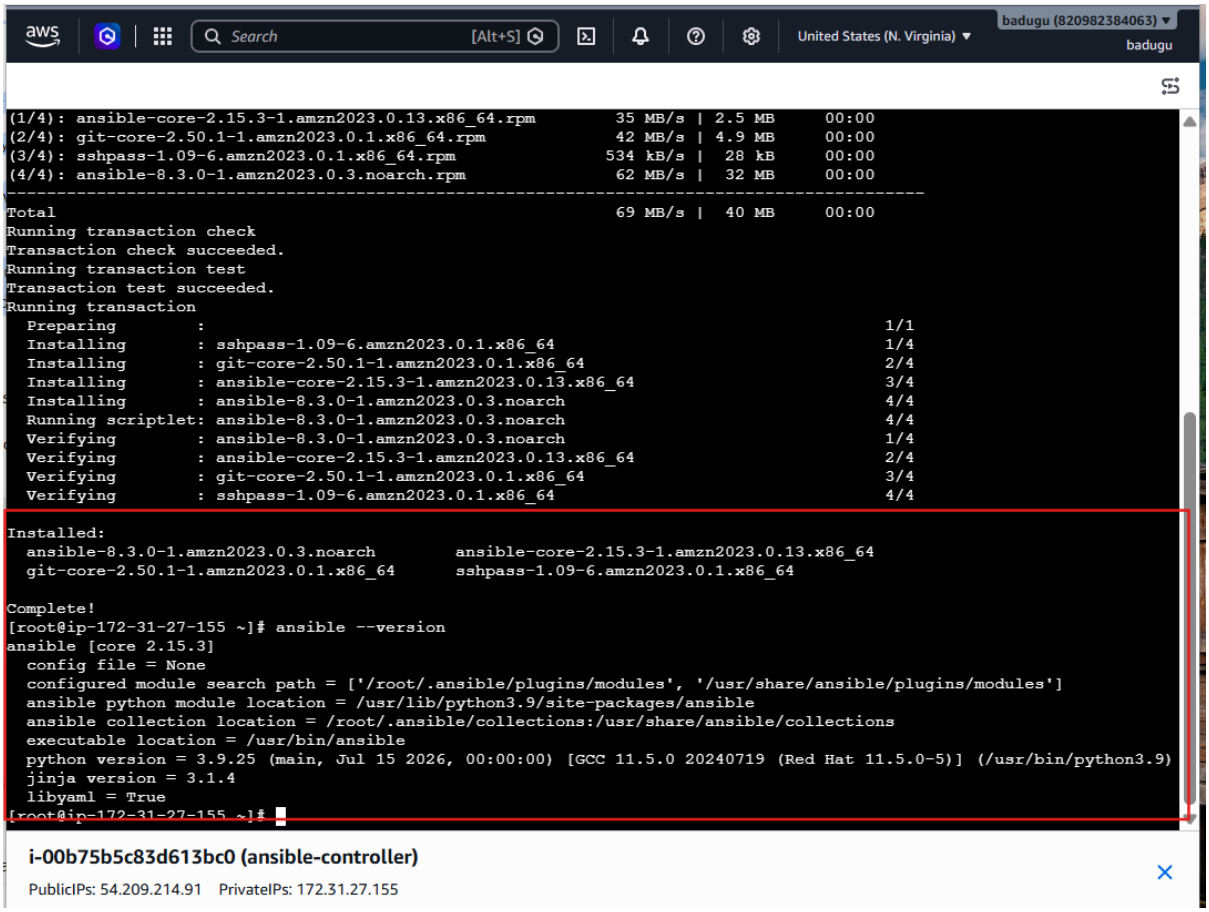
Updates the package lists.

```
ubuntu@ip-172-31-38-97:~$ sudo apt update
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:4 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [602 kB]
Get:7 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [138 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [4616 B]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [202 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [65.4 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55.6 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552 B]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64v3 Packages [498 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [99.6 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64v3 Packages [20.1 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [3728 B]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Get:21 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
Get:22 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:23 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64v3 Packages [291 kB]
Get:24 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
Get:25 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 Components [50.0 kB]
Get:26 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 c-n-f Metadata [8276 B]
Get:27 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [738 kB]
Get:28 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [173 kB]
Get:29 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [110 kB]
Get:30 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [5756 B]
Get:31 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [329 kB]
Get:32 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [107 kB]
Get:33 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:34 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [4704 B]
Get:35 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64v3 Packages [508 kB]
Get:36 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [101 kB]
Get:37 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64v3 Packages [21.1 kB]
Get:38 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [4168 B]
Get:39 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Components [216 B]
Get:40 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 c-n-f Metadata [256 B]
Get:41 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/main amd64 Components [212 B]
Get:42 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/main amd64 c-n-f Metadata [112 B]
Get:43 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe amd64v3 Packages [3144 B]
Get:44 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe Translation-en [7988 B]
Get:45 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Get:46 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 c-n-f Metadata [116 B]
Get:47 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 Components [216 B]
Get:48 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 c-n-f Metadata [120 B]
Get:49 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 Components [216 B]
```

Install Ansible.



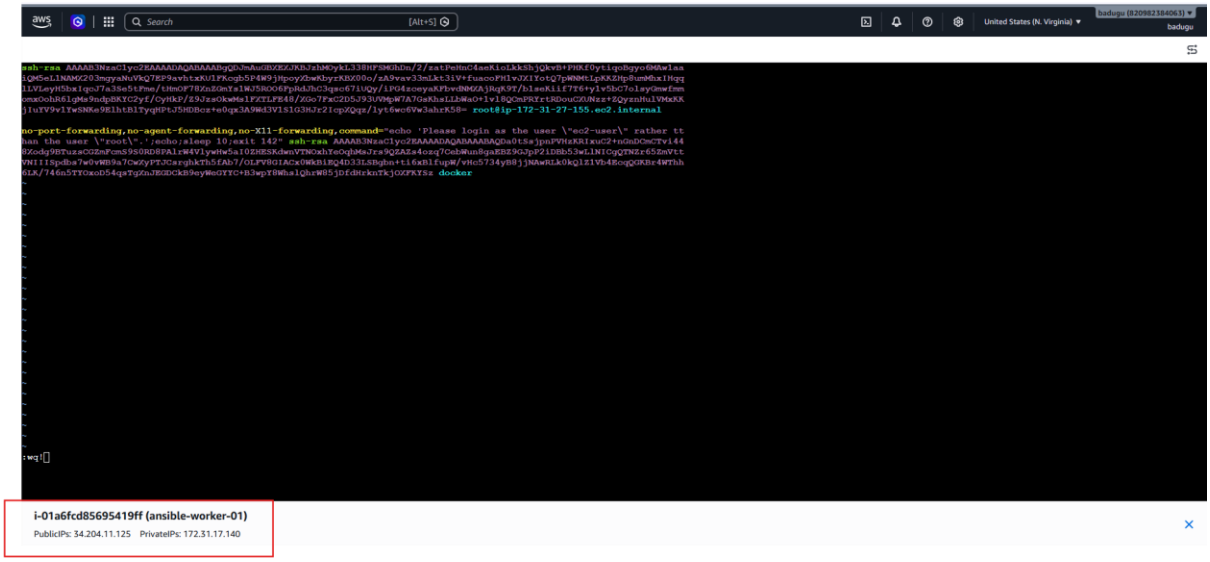
Verifies the installation.



Step 3: Prepare the two worker nodes

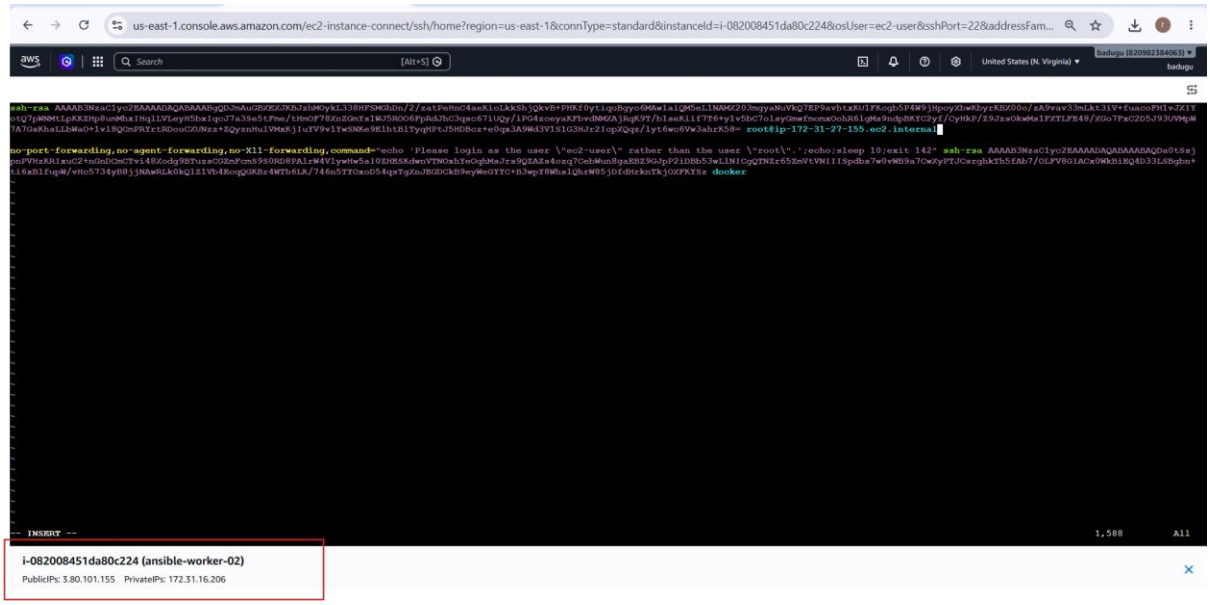
Connect to each worker separately and update packages.
Ansible normally uses SSH and Python to manage Ubuntu workers.

On worker 1:



On worker 2:

```
ubuntu@ip-172-31-43-183:~$ sudo apt update
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:5 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [602 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [138 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [4616 B]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [202 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [65.4 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55.6 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552 B]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64v3 Packages [498 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [99.6 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64v3 Packages [20.1 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [3728 B]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Get:20 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:21 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
Get:22 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:23 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64v3 Packages [291 kB]
Get:24 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
```

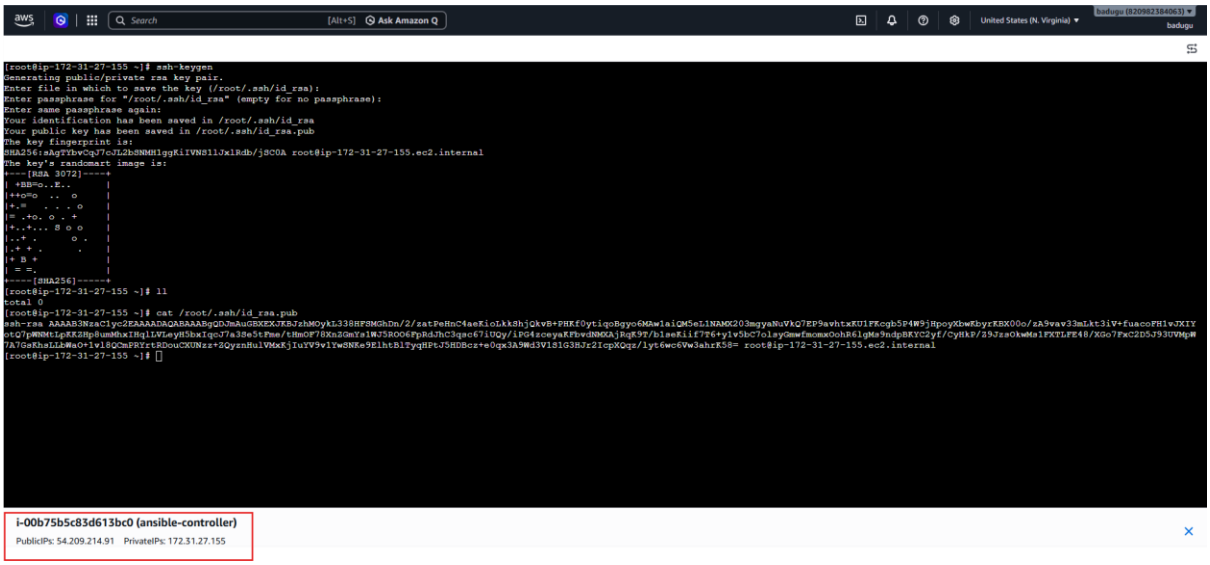


Step 4: Configure SSH from Master to Workers

Ansible must be able to SSH from the master to both workers without interactive password prompts.

On the master, create an SSH key:

Press Enter to accept the default file location. Leave the passphrase empty.



Then display the public key

```
ubuntu@ip-172-31-38-97:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAILye7N6gVM8SQ1Vm5ZohKGzmkR9AdKD6oa7nxHmPQ7Pe ubuntu@ip-172-31-38-97
ubuntu@ip-172-31-38-97:~$
```

Copy the displayed public key.

On each worker, add the master's public key to the ubuntu user's authorized keys:

On worker 1:

Create a directory, give permissions and add the public key to authorized keys file.

```
ubuntu@ip-172-31-34-95:~$ mkdir -p ~/.ssh
chmod 700 ~/.ssh
vi ~/.ssh/authorized_keys
ubuntu@ip-172-31-34-95:~$ nano ~/.ssh/authorized_keys
ubuntu@ip-172-31-34-95:~$
```

Added the public key to authorized keys file using nano editor.

To save press Ctrl+O, press Enter, and exit with Ctrl+X

On worker 2:

To save press Ctrl+O, press Enter, and exit with Ctrl+X

To avoid copying your private SSH key to the master set permissions to 600 on both workers.

Step 5: Configure the Ansible inventory

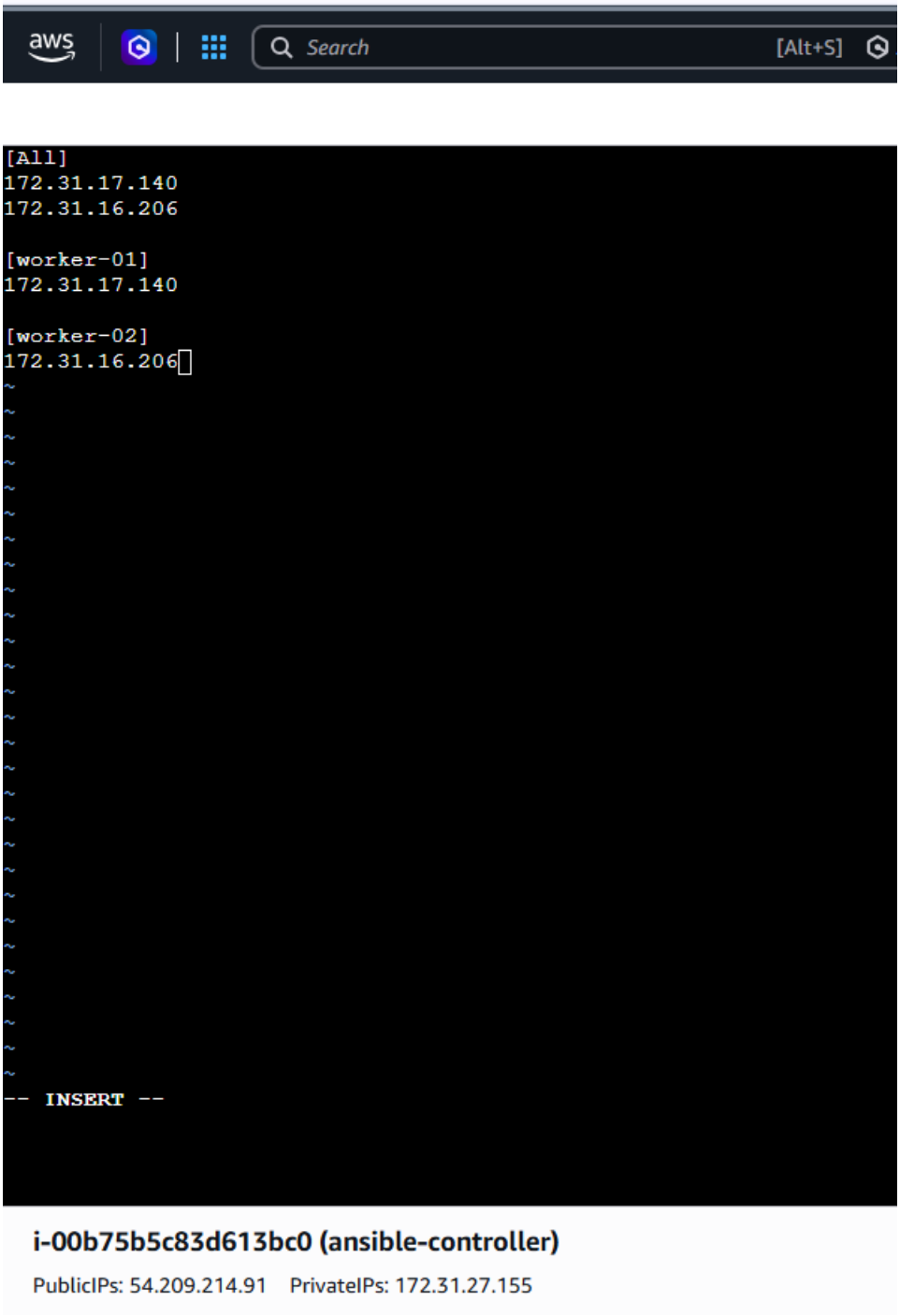
On master:

Create a directory to store Ansible configuration and inventory files.

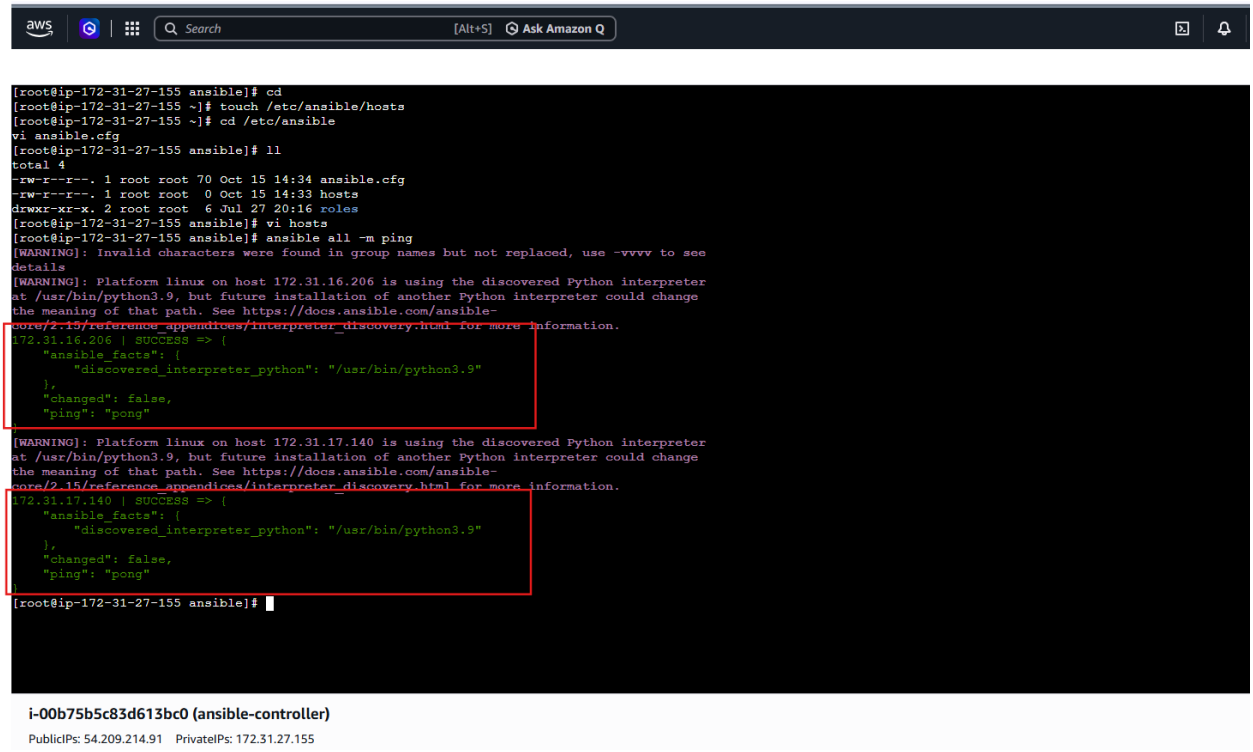
Add Ansible-managed servers in the inventory file using nano editor

Added Ip addresses of the worker nodes in inventory file.

Now to save press Ctrl+O, press Enter, and exit with Ctrl+X



Step 6: Test connectivity

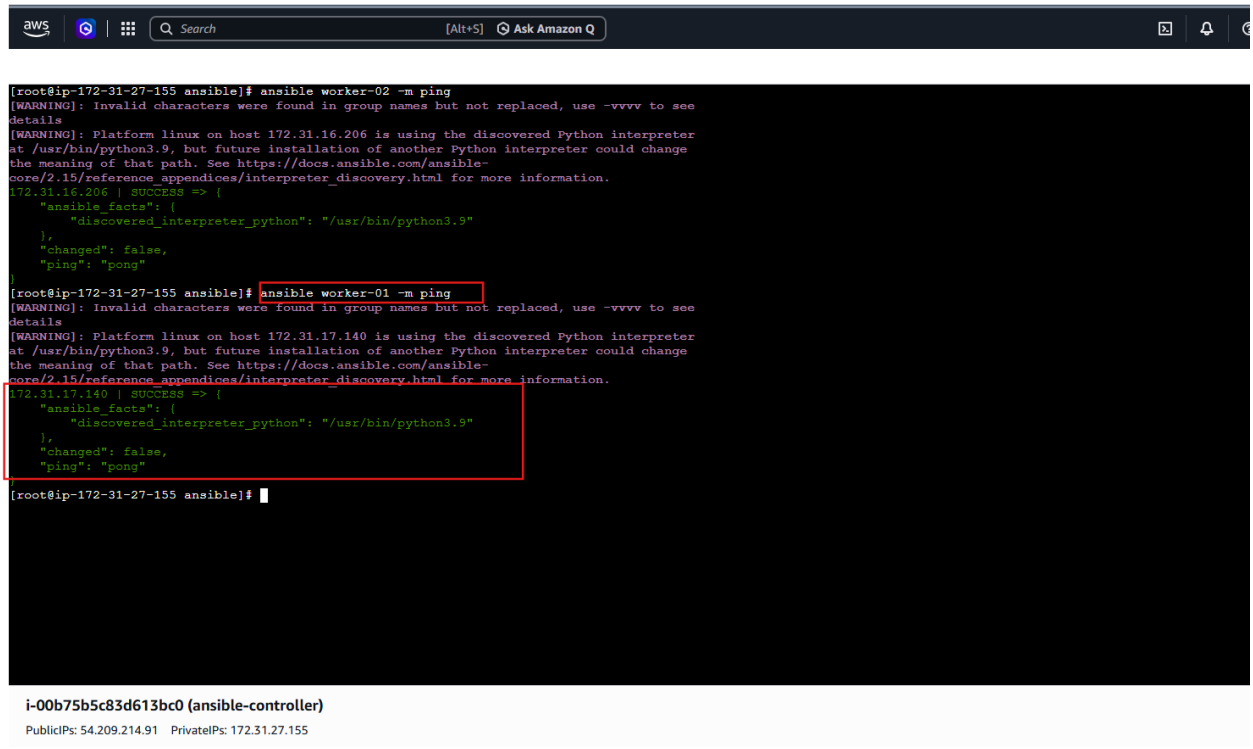


As both workers return pong, Ansible master-worker setup is working.n

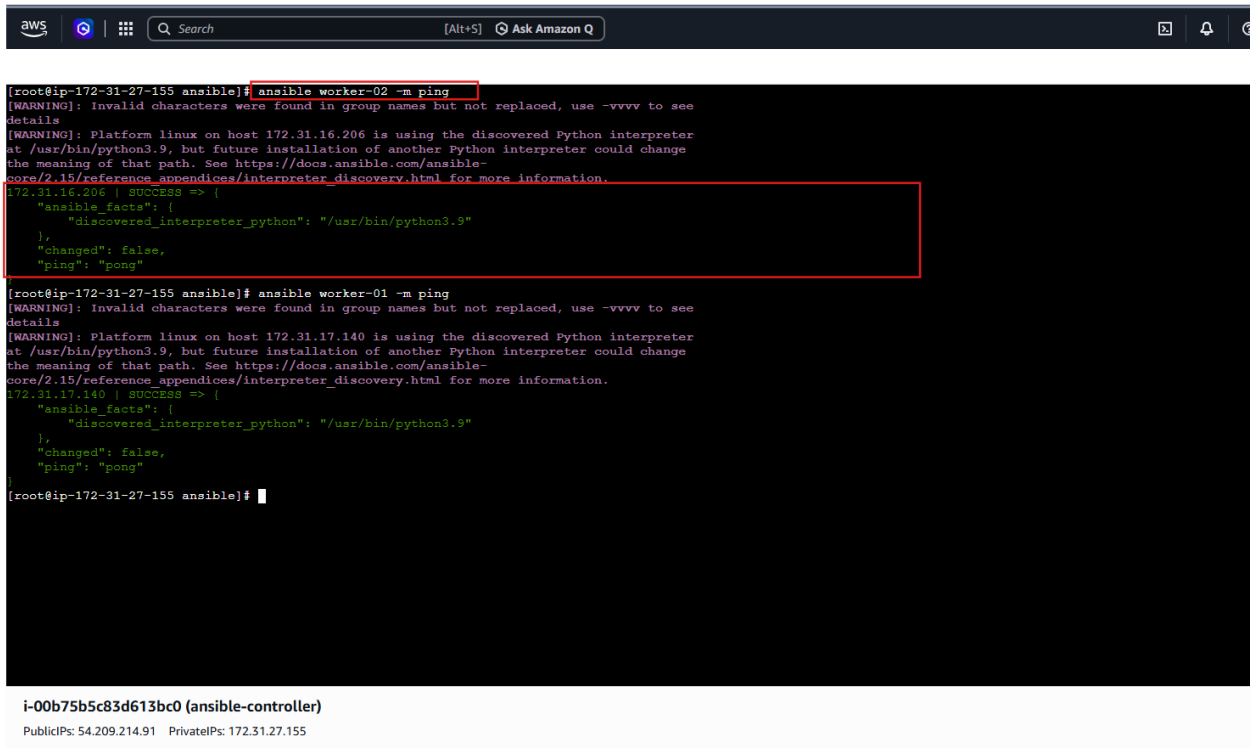
2. Execute the adhoc command shared in #dvps-cloud-documents-12 Channel.

1: Test connectivity

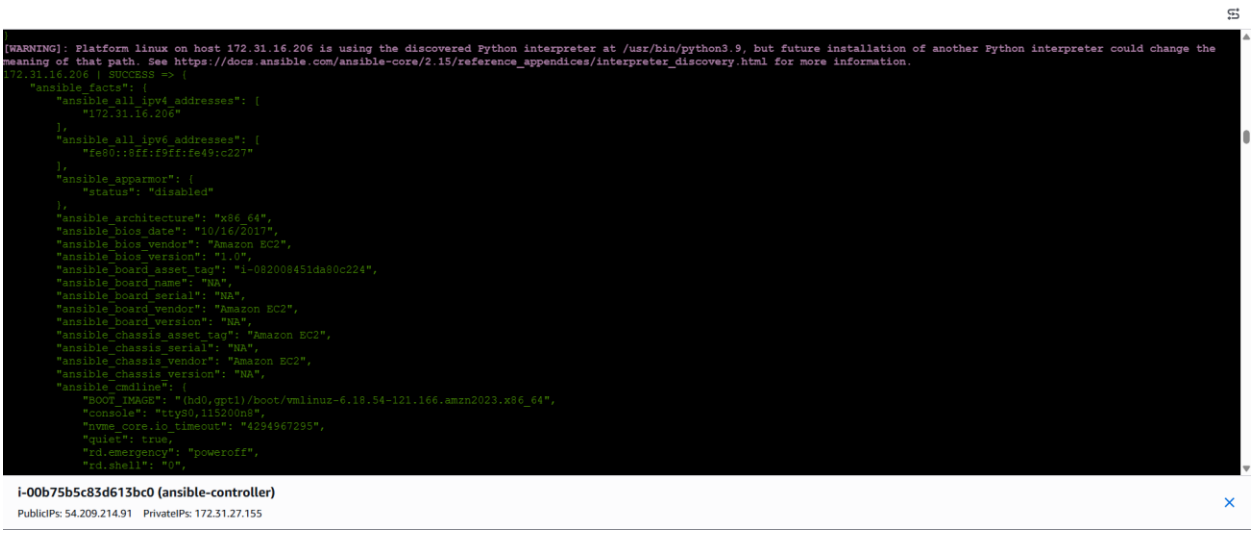
Target only worker 01:



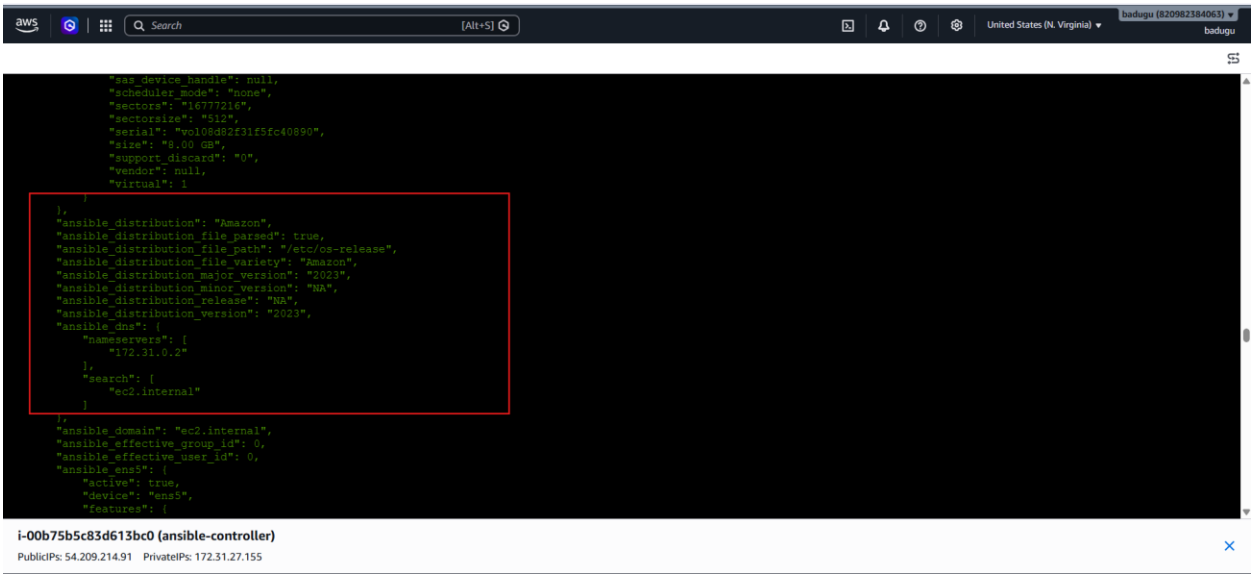
Target only worker 02:



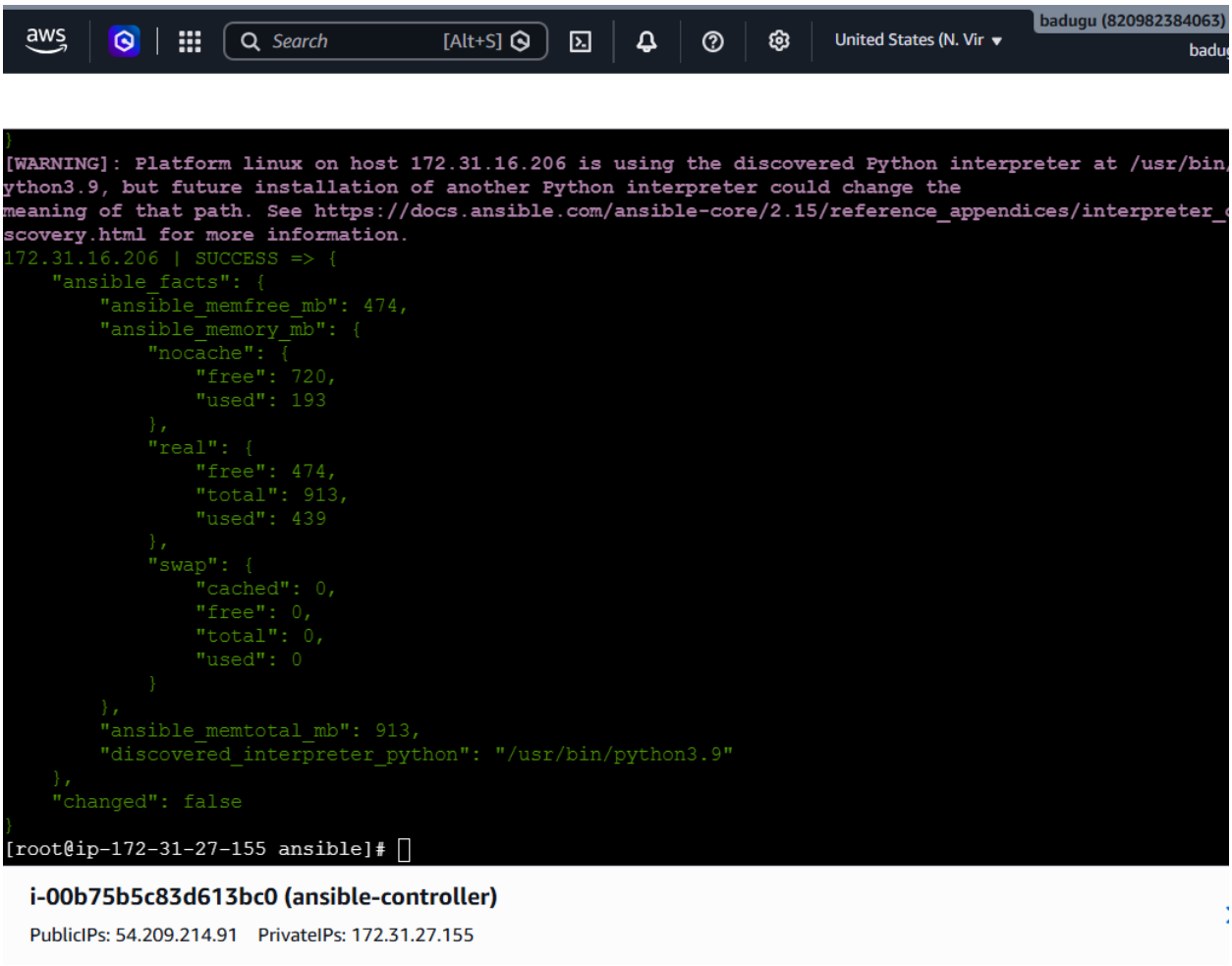
2: Gather server information/Gather facts of slave machine



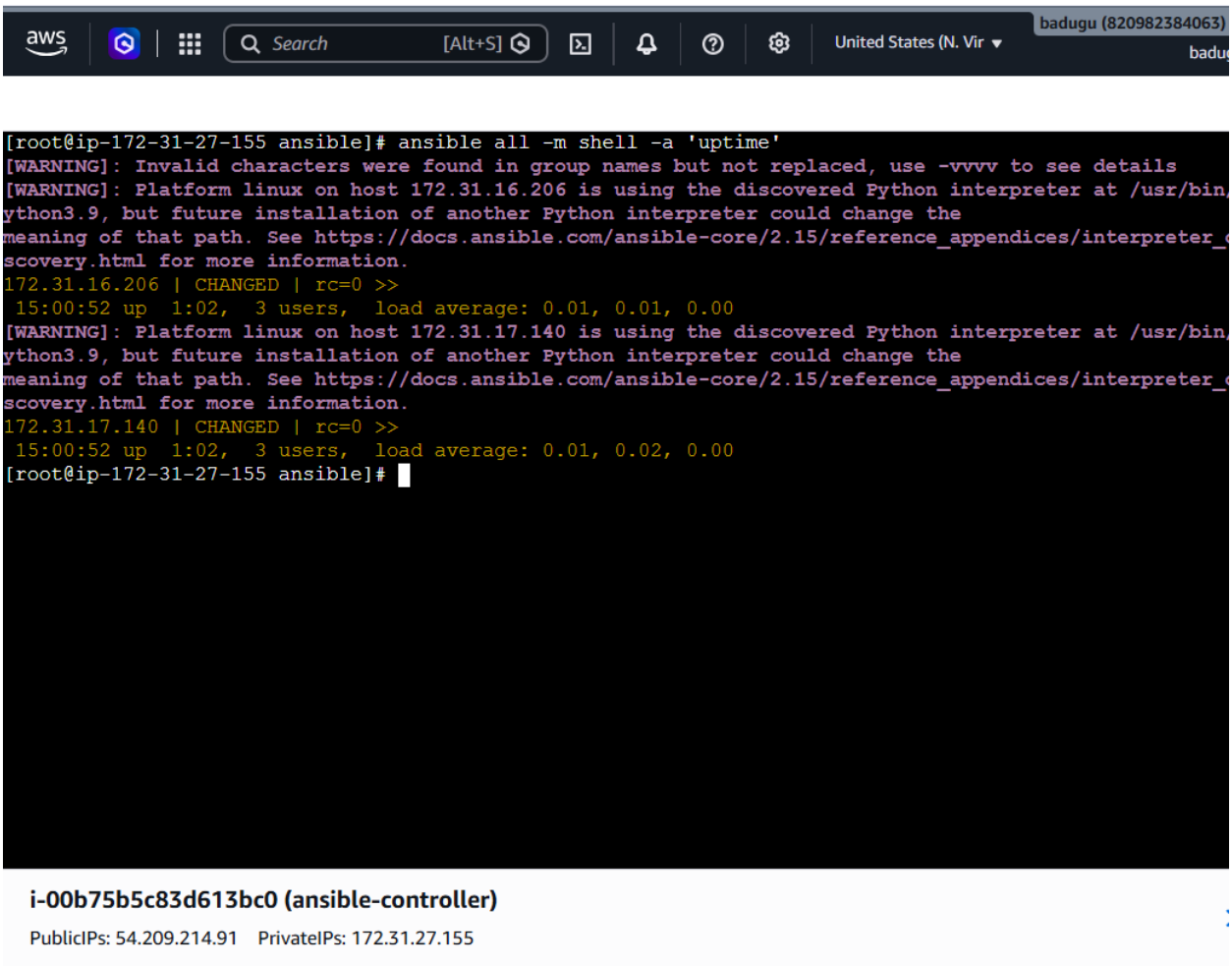
Display only operating system distribution information:



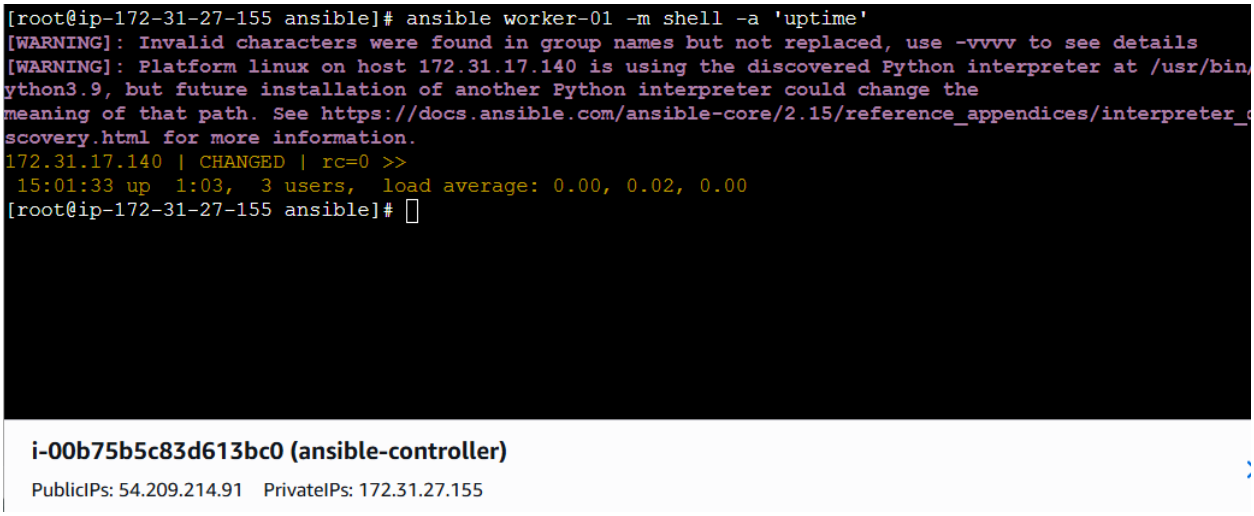
Display memory facts:



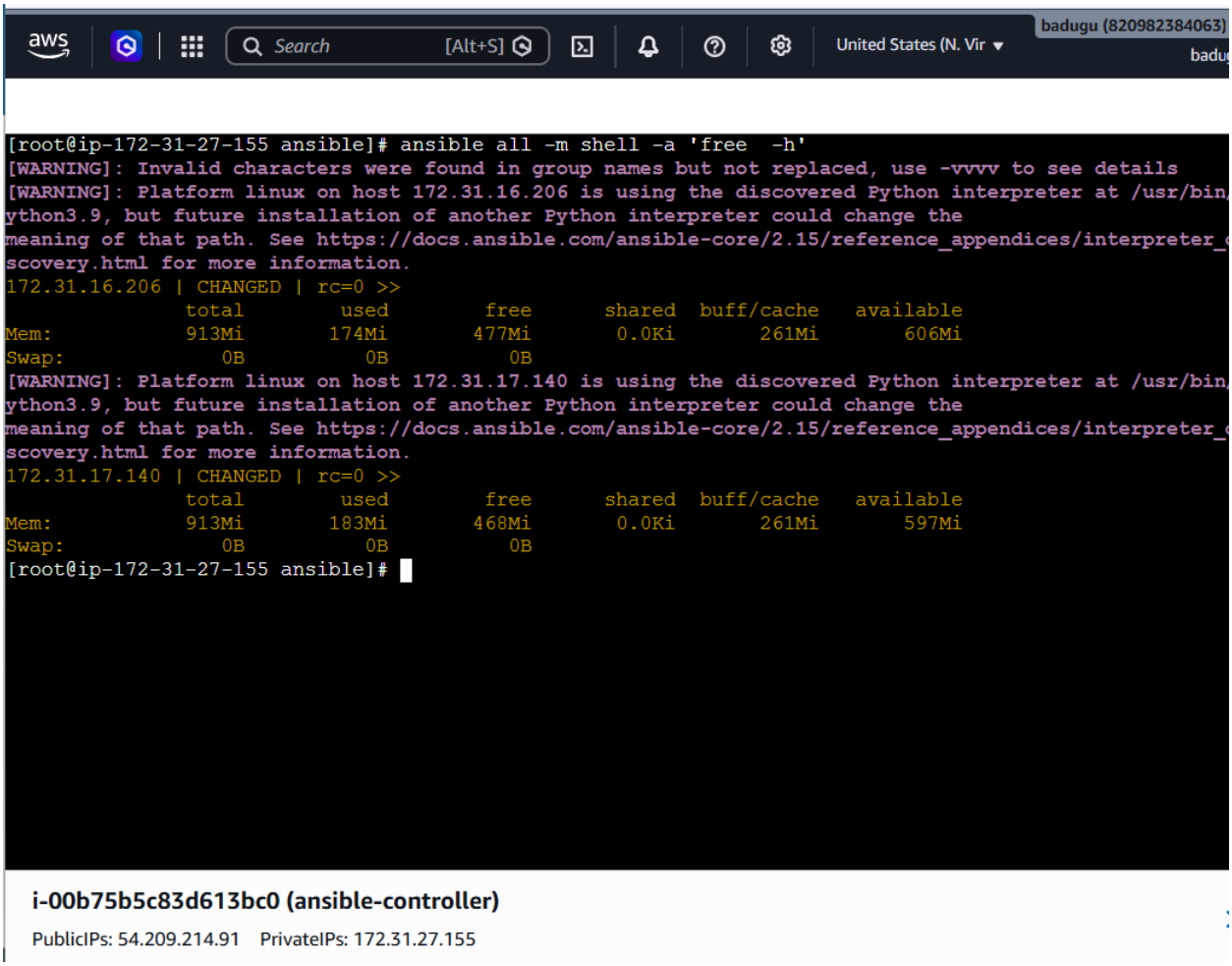
3. Check uptime



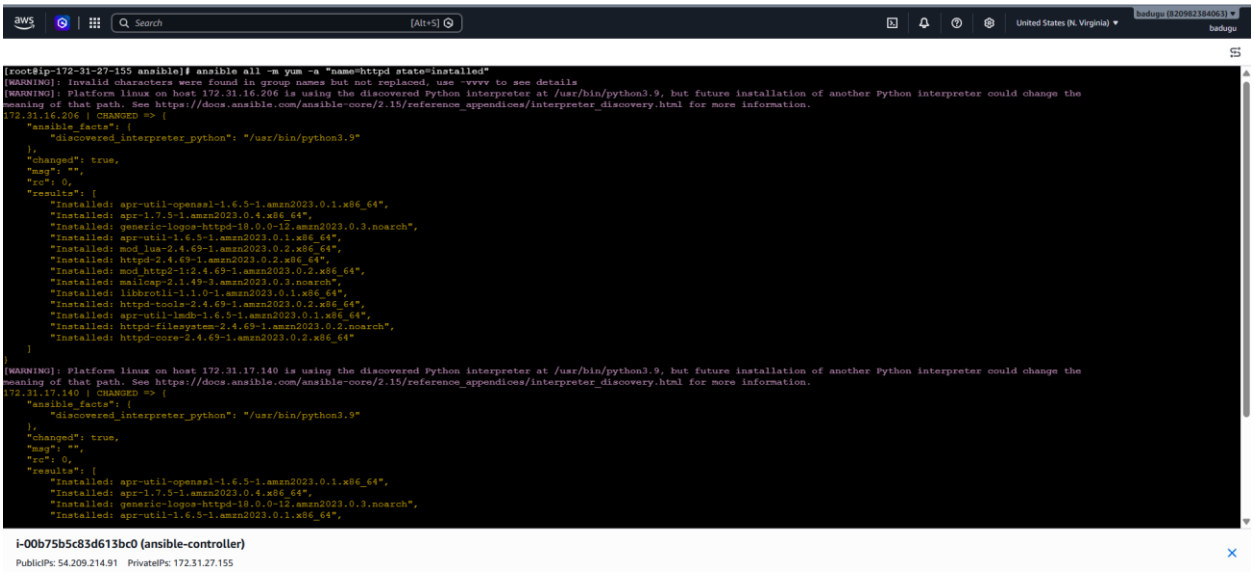
Target one worker:



4. Check free memory



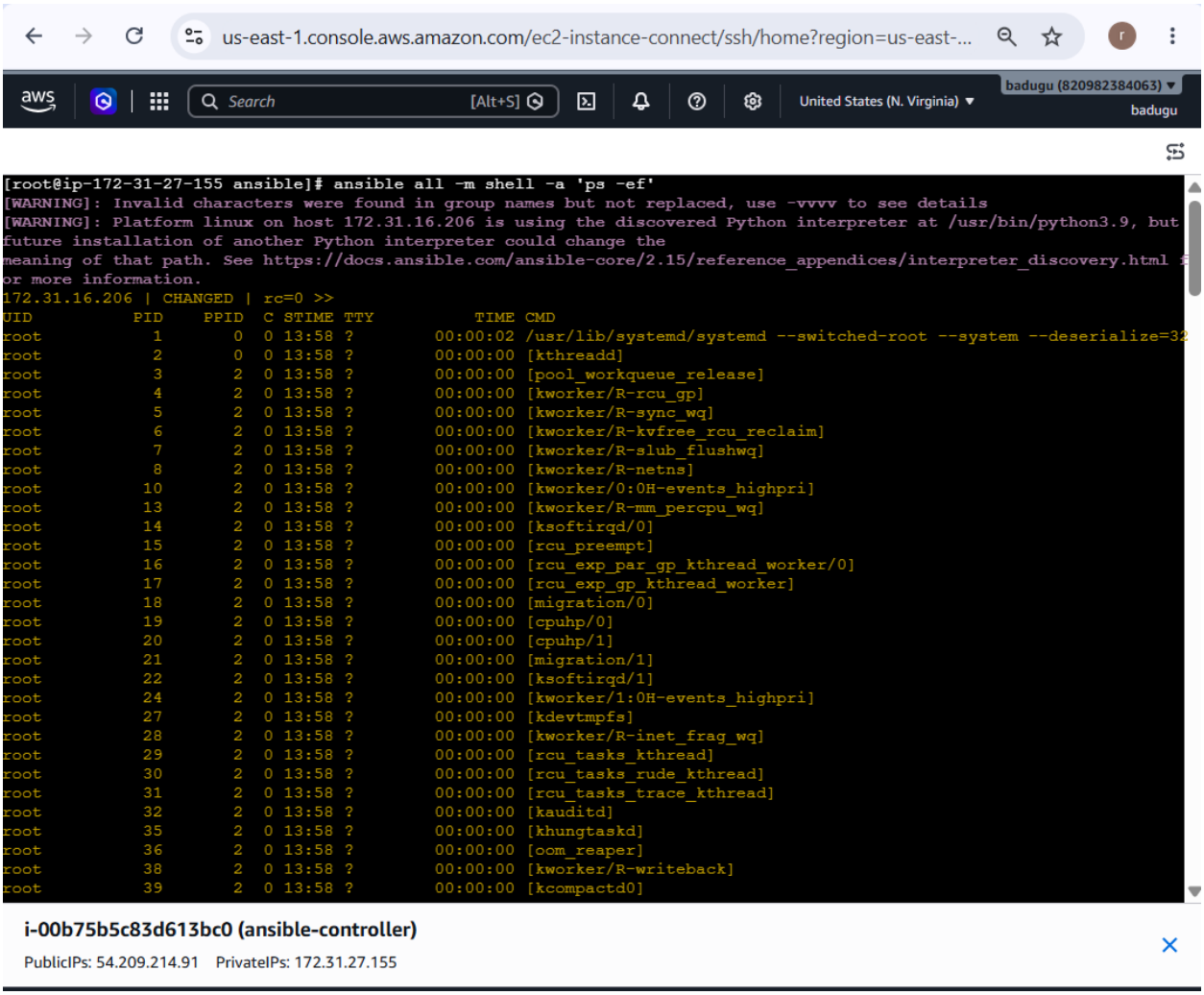
5. Check running processes



aws
Search
[Alt+S]
United States (N. Virginia)
badugu (820982384063)
badugu

```
[root@ip-172-31-27-155 ansible]# ansible all -m service -a "name=httpd state=started enabled=yes"
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details
[WARNING]: Platform linux on host 172.31.16.206 is using the discovered Python interpreter at /usr/bin/python3.9, but
future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html f
or more information.
172.31.16.206 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "enabled": true,
    "name": "httpd",
    "state": "started",
    "status": {
        "AccessSELinuxContext": "system_u:object_r:httpd_unit_file_t:s0",
        "ActiveEnterTimestampMonotonic": "0",
        "ActiveExitTimestampMonotonic": "0",
        "ActiveState": "inactive",
        "After": "remote-fs.target basic.target sysinit.target -.mount httpd-init.service network.target systemd-journ
ald.socket system.slice systemd-tmpfiles-setup.service nss-lookup.target tmp.mount",
        "AllowIsolate": "no",
        "AssertResult": "no",
        "AssertTimestampMonotonic": "0",
        "Before": "shutdown.target",
        "BlockIOAccounting": "no",
        "BlockIOWeight": "[not set]",
        "CPUAccounting": "yes",
        "CPUAffinityFromNUMA": "no",
        "CPUQuotaPerSecUSec": "infinity",
        "CPUQuotaPeriodUSec": "infinity",
        "CPUSchedulingPolicy": "0",
        "CPUSchedulingPriority": "0",
        "CPUSchedulingResetOnFork": "no",
        "CPUShares": "[not set]",
        "CPUUsageNSec": "[not set]",
        "CPUWeight": "[not set]",
        "CacheDirectoryMode": "0755",
```

i-00b75b5c83d613bc0 (ansible-controller)
PublicIPs: 54.209.214.91 PrivateIPs: 172.31.27.155



6. Create a directory

Below used Ansible command creates a directory named controller on all hosts listed your Ansible inventory.

```
ubuntu@ip-172-31-38-97:~$ ansible all -m file -a 'path=/home/ubuntu/controller state=directory'
worker2 | CHANGED => {
    "changed": true,
    "gid": 1000,
    "group": "ubuntu",
    "mode": "0775",
    "owner": "ubuntu",
    "path": "/home/ubuntu/controller",
    "size": 4096,
    "state": "directory",
    "uid": 1000
}
worker1 | CHANGED => {
    "changed": true,
    "gid": 1000,
    "group": "ubuntu",
    "mode": "0775",
    "owner": "ubuntu",
    "path": "/home/ubuntu/controller",
    "size": 4096,
    "state": "directory",
    "uid": 1000
}
ubuntu@ip-172-31-38-97:~$
```

Verify it the directory exists:

```
ubuntu@ip-172-31-38-97:~$ ansible all -m shell -a 'ls -ld /home/ubuntu/controller'
worker2 | CHANGED | rc=0 >>
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 15 09:17 /home/ubuntu/controller
worker1 | CHANGED | rc=0 >>
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 15 09:17 /home/ubuntu/controller
ubuntu@ip-172-31-38-97:~$
```

7. Install Apache on Ubuntu

Update package lists on both workers:

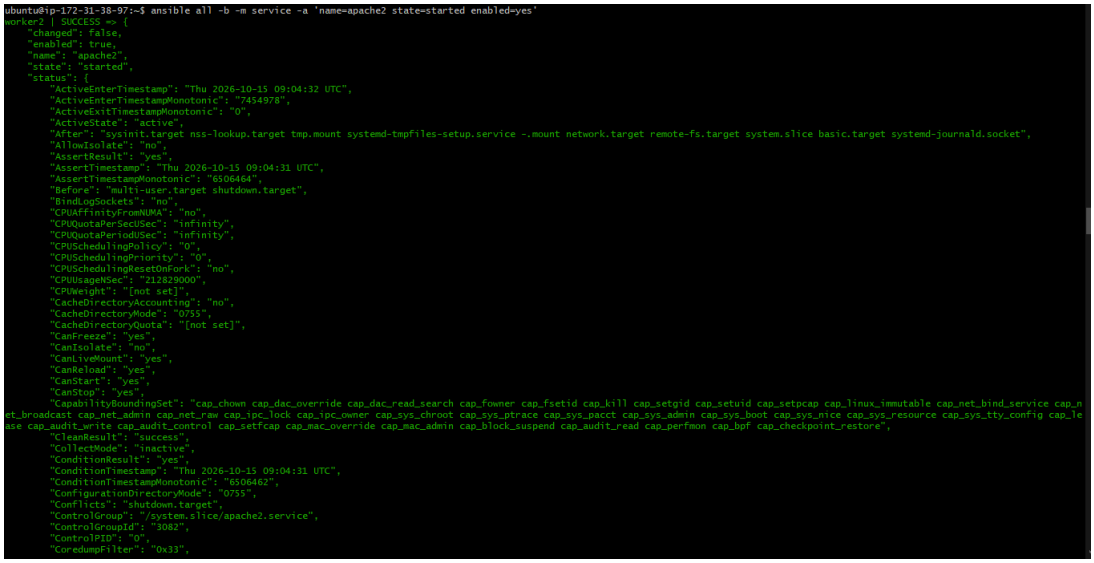
```
ubuntu@ip-172-31-38-97:~$ ansible all -b -m apt -a 'update_cache=yes'
worker2 | CHANGED => {
    "cache_update_time": 1792056197,
    "cache_updated": true,
    "changed": true
}
worker1 | CHANGED => {
    "cache_update_time": 1792056197,
    "cache_updated": true,
    "changed": true
}
ubuntu@ip-172-31-38-97:~$
```

Install Apache:

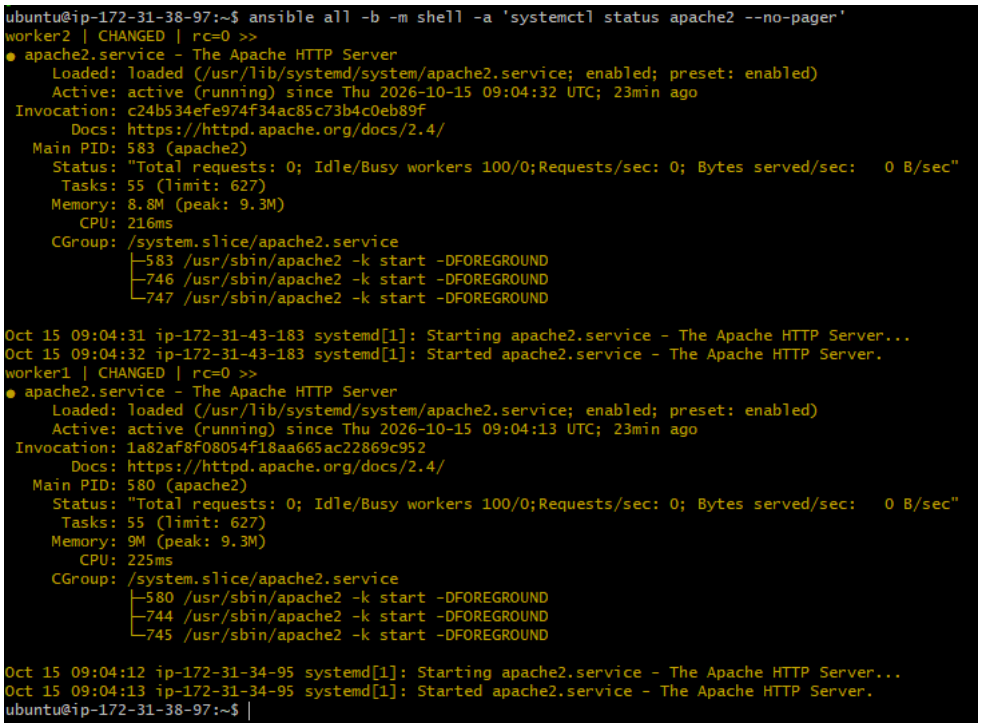
This command installs Apache2 on all worker nodes using the Ubuntu package manager.

```
ubuntu@ip-172-31-38-97:~$ ansible all -b -m apt -a 'name=apache2 state=present'
worker2 | SUCCESS => {
    "cache_update_time": 1792056197,
    "cache_updated": false,
    "changed": false
}
worker1 | SUCCESS => {
    "cache_update_time": 1792056197,
    "cache_updated": false,
    "changed": false
}
```

Start Apache and enable it at boot:

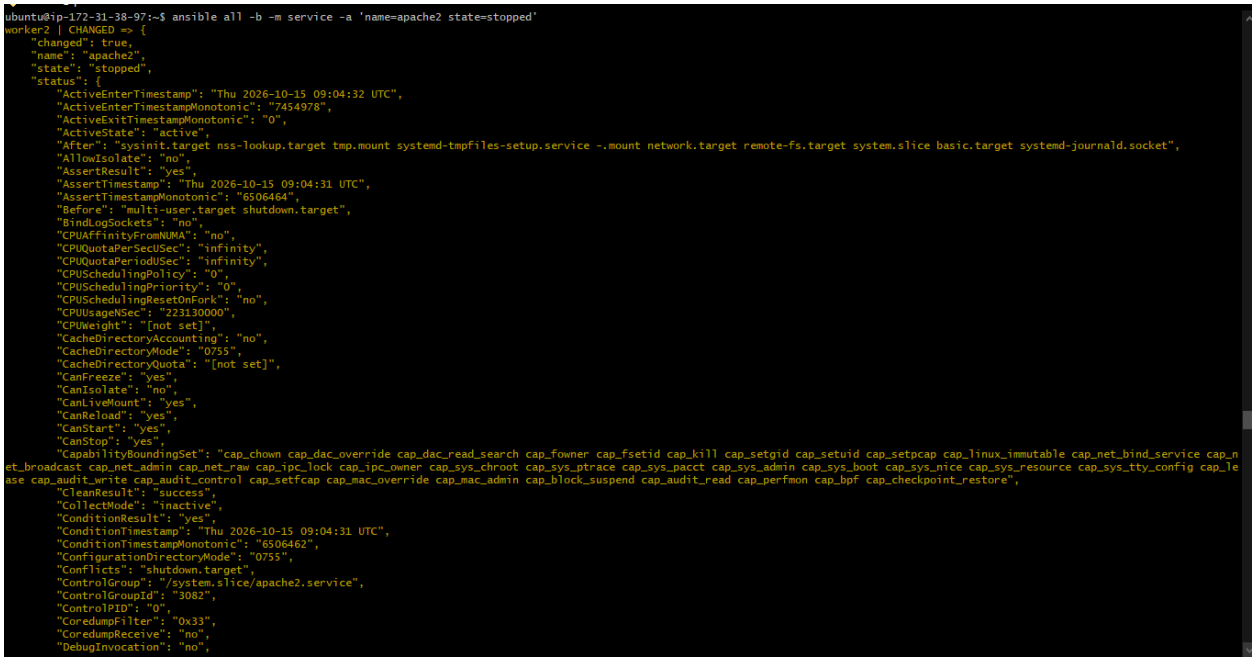


Check Apache's status:



Here, -b in the command means become root using privilege escalation. SSH user must have permission to use sudo.

8. Stop Apache



9. Uninstall Apache

This removes Apache from both workers.



10. Display detailed execution logs

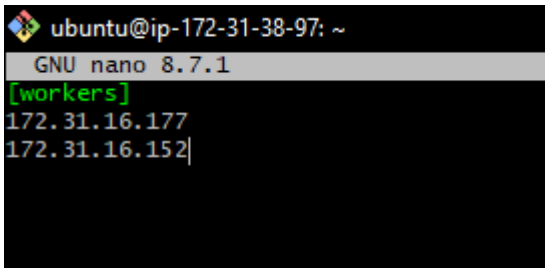
-vvvv displays detailed debugging information, useful when troubleshooting SSH or connectivity problems.





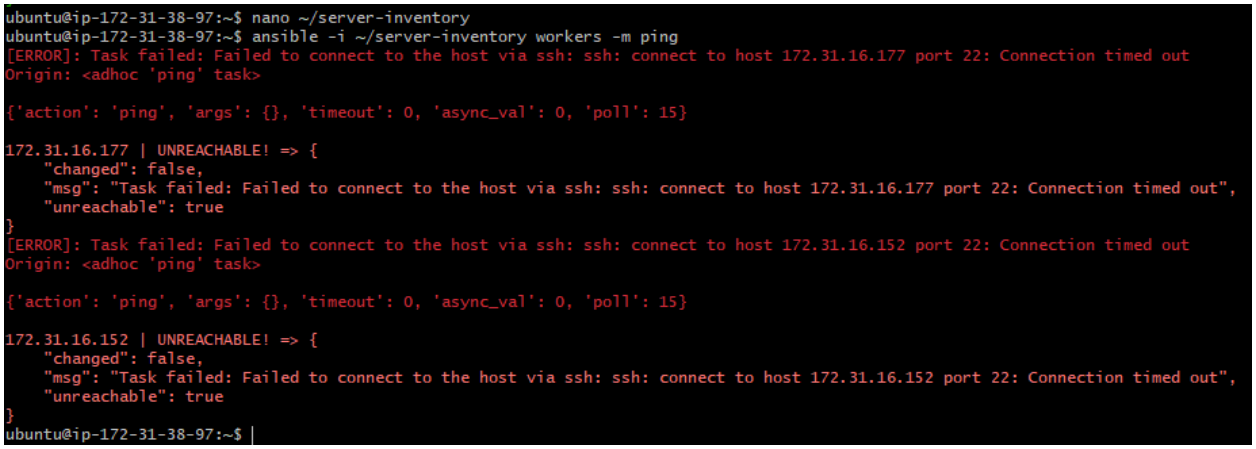
11. Use a custom inventory file

Create a separate inventory:



Test SSH connectivity between Ansible master and the hosts in the workers group using a specific inventory file that is newly created.

The -i option specifies the inventory file, while workers selects the group defined inside it.
It says unreachable as we there are no servers with those Ip's created.



12. Use username and password authentication

If we have not configured SSH keys, you can ask Ansible to prompt for an SSH password:

